

Dr Annie Holtz

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Research Interests

Cognitive biases, language development, phonology, systematic structure, cultural evolution & linguistic typology

Employment

The University of Edinburgh

POSTDOCTORAL RESEARCHER

- Pls: Prof. Mitsuhiro Ota & Prof. Kenny Smith
- ESRC Project: *Language learning, communication and the emergence of phonotactic constraints*

EXTENDED LEAVE DUE TO INJURY

- Worked reduced hours between June 2025 – August 2025
- On leave August 2025 – November 2025

Edinburgh, Scotland

Mar. 2024 – current

Jun. 2025 – Nov. 2025

The University of Edinburgh

RESEARCH ASSISTANT - CENTRE FOR LANGUAGE EVOLUTION

- Pl: Prof. Jennifer Culbertson
- ESRC project: *Connecting cognitive biases and typological universals in syntax*

Edinburgh, Scotland

Apr. 2019 – May 2019

The University of Edinburgh

RESEARCH ASSISTANT (VOLUNTEER POSITION) - EDINBURGH LABORATORY FOR LANGUAGE DEVELOPMENT

- Pls: Prof. Mitsuhiro Ota & Dr. Barbora Skarabela
- ESRC project: *The role of baby-talk words in early language development*

Edinburgh, Scotland

Jan. 2017 – Apr. 2019

Education

The University of Edinburgh

PHD IN LINGUISTICS AND ENGLISH LANGUAGE

- Supervisors: Prof. Jennifer Culbertson and Prof. Simon Kirby
- Thesis: *Cognitive biases in competition: Innovation and the evolution of language structure*

Edinburgh, Scotland

Sep. 2019 – Mar. 2024

The University of Edinburgh

MA(HONS) ENGLISH LANGUAGE AND LINGUISTICS

- Awarded: First-Class Honours degree
- Dissertation: *An experimental investigation of harmony and isomorphism in the noun phrase*

Edinburgh, Scotland

Sep. 2015 – Jul. 2019

Publications and manuscripts

Holtz, A., Kirby, S., & Culbertson, J. (invited revision). [Does degree of innovation modulate the influence of cognitive biases on language structure?](#) *Language*.

Holtz, A., Culbertson, J., & Kirby, S. (2026). [Category-specific and system-wide preferences in competition: Evidence from noun phrase harmony](#). *Glossa: Psycholinguistics* 5(1).

Holtz, A., Kirby, S., & Culbertson, J. (2025). [With or without a system: How category-specific and system-wide cognitive biases shape word order](#). *Cognitive Science* 49: e70139.

Holtz, A., Smith, K., & Ota, M. (2025). [The impact of similar place avoidance on novel word learning in adults](#). *Proceedings of the Annual Meeting of the Cognitive Science Society*, 47. Cognitive Science Society.

Dobrev, R., Birch, A., Keller, F., & **Holtz, A.** (2025). [Path encoding and manner salience in motion event descriptions: The case of Bulgarian and English](#). *Proceedings of the Annual Meeting of the Cognitive Science Society*, 47. Cognitive Science Society.

Holtz, A., & Papineau, B. (2023). [Examining the female-talker default in experimental language acquisition research](#). *Journal of Infant and Child Development*, 33(1), e2475.

Holtz, A., Kirby, S., & Culbertson, J. (2022). [The influence of category-specific and system-wide preferences on cross-linguistic word order patterns](#). In *Proceedings of the Annual Meeting of the Cognitive Science Society*, 44. Cognitive Science Society.

Holtz, A. (2021). [Ambiguity and variable phonological rules: The challenge of TD deletion in US English](#). In **A. Holtz, I. Kovač, R. Puggaard-Rode, & J. Wall** (Eds.), *ConSOLE XXIX: Proceedings of the 29th Conference of the Student Organization of Linguistics in Europe*. Leiden University Centre for Linguistics.

Martin, A., **Holtz, A.**, Abels, K., Adger, D., & Culbertson, J. (2020). [Experimental evidence for the influence of structure and meaning on linear order in the noun phrase](#). *Glossa*, 5(1), 1–21.

Invited Talks

Holtz, A. (12th March 2026) *It depends on what you know: System knowledge and competing pressures on language structure* [Invited talk]. The Linguistics Circle, Edinburgh University, UK.

Holtz, A. (26th June 2025) *Everything changes: How stages of language evolution activate different linguistic biases* [Invited talk]. National Science Foundation workshop on abstract and item-specific knowledge across domains and frameworks. Eugene, OR.

Holtz, A. (19th June 2025) *Biased minds, clustered languages: How cognition shapes linguistic diversity* [Invited talk]. Mechanisms of Social Behavior Research Group at Karolinska Institute. Stockholm, Sweden.

Holtz, A. (11th Nov 2024) *Defaulting to women's voices: The metascience of speaker choice in experimental language acquisition research* [Invited talk]. Edinburgh University Linguistics Society. Edinburgh, UK.

Holtz, A., Meisezahl, M. and Tal, S. (21st Oct 2022) *Introduction to artificial language learning experiments* [Invited workshop talk]. University of Konstanz. Konstanz, Germany.

Selected Conference Contributions

Holtz, A., Smith, K., & Ota, M. (2025). *The impact of similar place avoidance on novel word learning in adults* [Poster presentation]. Annual Meeting of the Cognitive Science Society, 47. San Francisco, USA.

Dobрева, R., Birch, A., Keller, F. & **Holtz, A.** (2025). *Path encoding and manner salience in motion event descriptions: the case of Bulgarian and English* [Poster presentation]. Annual Meeting of the Cognitive Science Society, 47. San Francisco, USA.

Zhou, R., Ding, W., **Holtz, A.**, Kirby, S., & Culbertson, J. (2024). *The Conceptual Origins of Adjective Ordering Preferences: Evidence from Silent Gesture* [Poster presentation]. Architectures and Mechanisms for Language Processing. Edinburgh, UK.

Holtz, A. (2024). *Using the Scale of Innovation to Study the Evolution of Language* [Oral presentation]. International Conference on the Evolution of Language. Madison, WI.

Holtz, A., Kirby, S., & Culbertson, J. (2022). *The effect of improvisation and learning on word order preferences* [Poster presentation]. Joint Conference on Language Evolution. Kanazawa, Japan.

Holtz, A., Kirby, S., & Culbertson, J. (2022). *The influence of category-specific and system-wide preferences on cross-linguistic word order patterns* [Oral presentation, online]. Annual Meeting of the Cognitive Science Society, 44. Toronto, Canada.

Holtz, A., Kirby, S., & Culbertson, J. (2022). *Cognitive biases differ in improvisation and learning of noun phrase word order* [Workshop presentation, online]. Workshop on Cognitive and Cultural Influences on Language Emergence. Istanbul, Turkey.

Holtz, A., Culbertson, J. and Kirby, S. (2022) *Postnominal adjectives and prenominal genitives as reflections of cognitive preferences* [Poster presentation, online]. Conference for the Student Organization of Linguistics in Europe, XXX. Nantes, France.

Holtz, A., Culbertson, J., Kirby, S. and Martin, A. (2021) *An experimental investigation of item-specific word order preferences* [Oral presentation, online]. Linguistics and English Language Postgraduate Conference). Edinburgh, UK.

Holtz, A. (2021). *Ambiguity and variable phonological rules: The challenge of TD Deletion in US English* [Poster presentation, online]. Conference for the Student Organization of Linguistics in Europe, XXIX. Leiden, Germany.

Ota, M., **Holtz, A.** and Skarabela, B. (2019) *Syllable repetition is privileged over consonant repetition in infant word segmentation* [Oral presentation]. Boston University Conference on Child Language Development. Boston, MA.

Holtz, A. (2019). *Cognitive biases and how they shape typology* [Oral presentation]. The 2019 conference of Undergraduate Linguistics Association of Britain. London, UK.

Editorials

Holtz, Annie, Iva Kovač, and Rasmus Puggaard-Rode eds. 2022. [ConSOLE XXX: Proceedings of the 30th conference of the student organization of linguistics in Europe](#) (25–27 January 2022, Nantes University). Leiden: Leiden University Centre for Linguistics.

Holtz, Annie, Iva Kovač, Rasmus Puggaard-Rode, and Joanna Wall, eds. 2021. [ConSOLE XXIX: Proceedings of the 29th conference of the student organization of linguistics in Europe](#) (26–28 January 2021, Leiden University). Leiden: Leiden University Centre for Linguistics.

Supervision

PhD **Radina Dobрева (co-supervised)** 2024 – current (passed viva Apr. 2026)
Masters **Ruizhe Zhou (co-supervised)** 2021 – 2022
Masters **Wei Ding (co-supervised)** 2021 – 2022

Edinburgh University
Edinburgh University
Edinburgh University

Teaching

Origins and Evolution of Language

Edinburgh University

GUEST LECTURER

Autumn 2024

- Advanced course introducing concepts from biology, cognitive science and linguistics and how they apply to the study of language
- Designed and delivered two lectures on how signing practices can inform our understanding of the evolution of language
- Designed one week of tutorial materials and subsequent essay assessment
- Course for Y3-Y4 UG and Masters students

First Language Acquisition

Edinburgh University

TUTOR/MARKER

Autumn 2023

- Advanced course on language acquisition, including acquisition of phonetics, phonology, morphology and syntax
- Ran three weekly tutorials, planned delivery of tutorial materials, organised study groups and marked essays and assignments
- Course for Y3-Y4 UG and Masters students

Simulating Language

Edinburgh University

TUTOR/MARKER

Spring 2021 & 2022

- Course teaching students Python programming and how to build agent-based Bayesian and iterated learning models
- Ran two weekly labs, planned delivery of lab materials and marked assignments
- Course for Y3-Y4 UG and Masters students

Linguistics and English Language 1A

Edinburgh University

TUTOR/MARKER

Autumn 2021

- Introductory course in linguistics covering communication, syntax, phonology, phonetics, morphology, semantics and discourse analysis
- Ran three weekly tutorials, planned delivery of tutorial materials, organised study groups and marked essays and assignments
- Course for Y1 UG students

LEL2D: Cross-linguistic Variation: Limits and Theories

Edinburgh University

MARKER AND SUBSTITUTE TUTOR

Spring 2020

- Course on typological variation, focusing on morphosyntax and phonology
- Supported teaching of tutorials and marking assignments when other tutors were absent
- Course for Y2 UG students

Skills Centre

Edinburgh University

TUTOR

Mar. 2020 – Dec. 2024

- JsPsych tutor doing one-on-one appointments with students to work on their code
- Academic writing tutor doing one-on-one appointments with students to work on their essay writing
- Programming tutor doing one-on-one appointments with students to work on their experiment code for theses and dissertations
- Available to all students

Online Experiments

Edinburgh University

TUTOR

Jan. 2020

- Short course on using JsPsych to programme online behavioural experiments
- Provide 1-to-1 programming support during labs
- Available to all students

Funding

Grant

Edinburgh University

ESRC FESTIVAL OF SOCIAL SCIENCE FUND

Jul. 2025

- Grant value: £500
- Funding to host public outreach event as part of ESRC Festival of Social Science 2026

Grant

Edinburgh University

PPLS STUDENT AND STAFF INITIATIVE FUND

2022 and 2023

- Grant value: £1,500 (2022) and £1,000 (2023)
- Funds used to organise annual training and community-building events for PhD students

Grant

Edinburgh University

DEPARTMENT OF LINGUISTICS AND ENGLISH LANGUAGE RESEARCH SUPPORT GRANT

2020 – 2023

- Grant value: £3,400
- Funds used for conference and workshop attendance during PhD, as well as payment for experiment participants

Scholarship

Edinburgh University

DEPARTMENT OF LINGUISTICS AND ENGLISH LANGUAGE PHD SCHOLARSHIP

2019 – 2023

- Stipend value: £12,981 (fees) and £43,659 (stipend)
- Covered fees and living costs for first 3 years of the PhD programme

Awards

2024 **Nominated** Students' Association Teaching Award – Student Tutor of the Year

Edinburgh University

2022 **Nominated** Students' Association Teaching Award – Student Tutor of the Year

Edinburgh University

Outreach

Language Evolves!

Edinburgh

DEMONSTRATOR

Autumn 2019, 2021

- Hosting online and in-person sessions introducing the concept of language evolution in primary schools
- Designing and delivering age-appropriate activities
- Total reach: 220 primary school students

UK Linguistics Olympiad

Edinburgh

DEMONSTRATOR

Autumn 2019

- Hosting practice sessions in schools that are preparing for the Linguistics Olympiad
- Promoting the Linguistics Olympiad in schools
- Total reach: 80 secondary school students

Professional Activities & Services

- Journal reviewer for *Journal of Memory and Language*, *Cognitive Science*, *Journal of Language Evolution*, *The Journal of the Undergraduate Linguistics Association of Britain* and *The ConSOLE Proceedings*
- Conference reviewer for *Association of Computational Linguistics (ACL)*, *CogSci*, *EvoLang*, *ProtoLang*, *ULAB Conference* and *ConSOLE*
- Board member for The Student Organization of Linguistics in Europe (2021-2023)
- Editor for *The ConSOLE Proceedings* (2021-2023)